Proper circular arc graphs are e-positive

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Abstract

We prove an e-positive formula for the chromatic symmetric function of proper circular arc graphs solving the $q = 1$ case of a conjecture of Ellzey and of Alexandersson and Panova. In doing so, we provide a new proof of the e-positivity of unit interval graphs, which alongside Guay-Paquet's reduction gives a new proof of the Stanley–Stembridge conjecture. We define color matrices, which count proper colorings, and tableau matrices, whose entries are nonnegative rational numbers and ratios of elementary symmetric functions. We prove the two matrices are related by a single family of change of basis matrices, which become invertible after restricting to finitely many colors, and we show the chromatic symmetric function of proper circular arc graphs comes from taking the trace of these matrices. Using Hikita's tableaux, this gives an explicit formula for the chromatic symmetric function of a proper circular arc graph as a weighted sum over tableaux whose first and last k vertices lie in the same columns.

1 Introduction

The chromatic symmetric function $X_G(\boldsymbol{x})$ generalizes the chromatic polynomial, encodes additional information about acyclic orientations, and has connections to Hessenberg varieties [5, 15] and Hecke algebras [6, 17]. The Stanley–Stembridge conjecture [18, Conjecture 5.1], [19, Conjecture 5.5] states that the chromatic symmetric function of the incomparability graph of a $(\mathbf{3}+\mathbf{1})$-free poset has nonnegative coefficients in the elementary symmetric function basis. By a reduction due to Guay-Paquet [9, Theorem 5.1], it suffices to consider unit interval graphs. Hikita [10, Theorem 1.6 and Corollary 1.8] resolved the conjecture by providing a formula in terms of probabilities associated to standard Young tableaux.

Shareshian and Wachs introduced the chromatic quasisymmetric function $X_G(\boldsymbol{x}; q)$, showed that it is symmetric for natural unit interval graphs, and conjectured that its e-coefficients are polynomials in q with nonnegative coefficients [16], which was recently proved [13, Theorem 1.1]. Ellzey generalized this function to directed graphs and conjectured the same positivity for proper circular arc digraphs [8, Conjecture 1.4], and the same conjecture, with unimodality, appears in [3, Conjecture 5.2]. At $q = 1$, this conjectures

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that every proper circular arc graph is e-positive. Several families built from cycles are known to be e-positive [4, 20, 23], but the general conjecture remains unsolved. For the related vertical-strip LLT polynomials, Alexandersson and Panova also conjectured that the circular versions are e-positive after the substitution $q \mapsto q+1$ [3, Conjecture 4.5], and this e-positivity was recently proved by Alexandersson and Dai [2, Theorem B].

The chromatic symmetric function is multiplicative over connected components, but gluing graphs at a vertex does not correspond to multiplication of their chromatic symmetric functions. In [21], Tom and the author gave a matrix formula for this operation by assigning an infinite e-positive matrix M_G, constructed from Hikita's tableau, to each natural unit interval graph. Matrix multiplication $M_G M_H = M_{G+H}$ corresponds to gluing two graphs together at a single vertex to form a larger natural unit interval graph, and the trace $\operatorname{tr} M_G = X_{G^\circ}(\boldsymbol{x})$ corresponds to identifying the first and last vertex of the graph to form a proper circular arc graph.

This paper generalizes the above result to multiple vertices. In particular, given a graph G we define color matrices, whose entries count proper colorings with specified colors on the marked vertices, and e-positive tableau matrices, which record the columns of specified boxes in Hikita's tableau. We show these matrices are related by a change of basis and satisfy the same multiplication and trace properties. Since every proper circular arc graph can be obtained by identifying the first and last k vertices of a natural unit interval graph, the tableau trace then proves that every proper circular arc graph is e-positive (Theorem 21), establishing the $q = 1$ case of Ellzey's conjecture.

Moreover, the tableau matrices give an explicit formula for the chromatic symmetric function of a proper circular arc graph (Theorem 1). We cut the circle to obtain a natural unit interval graph, place the first k vertices at the ends of rows of lengths $j_1, \ldots, j_k$, build the rest of the graph with Hikita's rule, and keep the tableaux whose last k vertices return to the columns $j_1, \ldots, j_k$. Unlike Hikita's formula, this sums over one starting tableau for each choice of j, so the number of terms grows exponentially in k, but every term is explicit. In relating the two matrices, we also provide a new proof of the Stanley–Stembridge conjecture.

More specifically, Section 2 provides definitions, Section 3 describes color matrices and their multiplication and trace properties, Section 4 defines the tableau operators and states the main theorem, which is proved in Section 5. Section 6 proves the trace formula for tableau matrices and establishes e-positivity. Finally, Section 7 describes the tableau matrices using Hikita's rule and proves the following explicit formula. The change-of-basis theorem, e-positivity results, and tableau formulas in this paper have been formalized in Lean 4 [7], using mathlib [11]. The source code, build instructions, and a correspondence between the statements of the paper and the Lean theorems are available at `github.com/goombamaui/ProperCircularArcGraphs`.

Theorem 1. *Let Γ be a proper circular arc graph on n vertices, obtained from a natural unit interval graph G on $[n+k]$ by identifying $n+r$ with r for $1 \le r \le k$ as in Lemma 6,*

and let $N \geq n$. Then

$$X_\Gamma(\boldsymbol{x}) = \sum_j \sum_{\substack{T \supseteq T_j \\ T \setminus T_j = \{k+1,\ldots,n+k\} \\ c_T(n+1,\ldots,n+k) = j}} \prod_{v=k+1}^{n+k} A\big(c_T(v) \mid c_T(N(v))\big) \frac{e_{c_T(v)}}{e_{c_T(v)-1}},$$

where for sequences σ of distinct positive integers,

$$A(1 \mid \sigma) = \prod_r \frac{\sigma_r - 1}{\sigma_r}, \qquad A(\sigma_r + 1 \mid \sigma) = \frac{\sigma_r + 1}{\sigma_r} \prod_{r' \neq r} \frac{\sigma_r - \sigma_{r'} + 1}{\sigma_r - \sigma_{r'}},$$

and otherwise $A(y \mid \sigma) = 0$, $e_0 = 1$, $N(v)$ is the sequence of neighbors of v in G smaller than v, and $c_T(v)$ is the column of v in T. The sum is over k-tuples j of distinct elements of $[N]$ and tableaux T with at most N columns, drawn with rows increasing to the right and columns increasing upward, where T_j has rows of lengths $j_1, \ldots, j_k$, with r at the end of the row of length j_r and distinct nonpositive entries elsewhere.

2 Definitions

Throughout, $N \in \mathbb{N} \cup \{\infty\}$ is the number of colors, with $[N] = \{1, \ldots, N\}$ for $N \in \mathbb{N}$ and $[\infty] = \mathbb{N}$. Let $\boldsymbol{x} = (x_c)_{c \in [N]}$ be commuting variables, one for each color, and e_j be the elementary symmetric functions in $\boldsymbol{x}$. In this section, all statements hold for every N.

For basis vectors, we suppress outer parentheses in tuple subscripts: $b_{a,w}$ means $b_{(a,w)}$, and likewise for $\hat{b}$. The empty tuple is denoted by $\emptyset$.

Definition 2. A *proper coloring* $\kappa : V(G) \to [N]$ assigns a number to each vertex so that adjacent vertices have different colors, meaning $\kappa(v) \neq \kappa(v')$ if $\{v, v'\} \in E(G)$. The *chromatic symmetric function* [18, Definition 2.1] is

$$X_G(\boldsymbol{x}) = \sum_{\kappa \text{ proper}} \prod_{v \in V(G)} x_{\kappa(v)},$$

where κ ranges over proper colorings. For finite N, $X_G(x_1, \ldots, x_N)$ is $X_G(\boldsymbol{x})$ with $x_c = 0$ for $c > N$, and when $N \geq |V(G)|$ this does not change the e-coefficients.

Definition 3. A graph G on $[n]$ is a *natural unit interval graph* (NUIG) if $\{i, j\} \in E(G)$ with $i < j$ implies $\{i, k\}, \{k, j\} \in E(G)$ for all $i < k < j$. Equivalently, a graph is isomorphic to a NUIG if and only if it is the intersection graph of finitely many intervals on the real line, none of which contains another [14]. A *proper circular arc graph* is then the intersection graph of finitely many arcs of a circle, none of which contains another.

Definition 4. A *marked graph* $\mathbf{G} = (G, \alpha, \beta)$ is a graph G with two sequences of distinct vertices α (the *outputs*) and β (the *inputs*); these are the bi-labeled graphs of [12, Definition 3.1]. If $\mathbf{H} = (H, \alpha', \beta')$ and β, α' have the same length, then $\mathbf{G} + \mathbf{H} = \big((G \sqcup H)/(\beta_r \sim \alpha'_r),\ \alpha,\ \beta'\big)$ [12, Definition 3.13]. If α, β have the same length, the *trace* $\operatorname{tr} \mathbf{G}$ is the graph $G/(\alpha_r \sim \beta_r)$.

Example 5. Figure 1 shows an example of $(G, (4, 5), (1, 2)) + (H, (3, 4), (1, 2))$ forming a new NUIG; note that duplicate edges do not change proper colorings so we can ignore them.

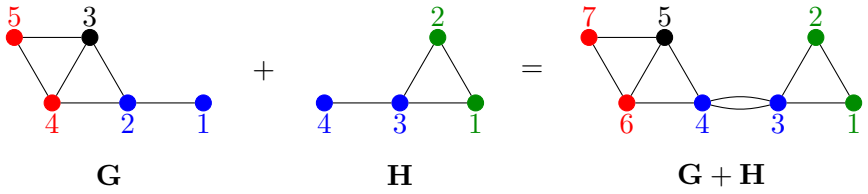


Figure 1: The sum $(G, (4, 5), (1, 2)) + (H, (3, 4), (1, 2))$.

Lemma 6. *Every proper circular arc graph is* $\operatorname{tr}\mathbf{G}$ *for some* $\mathbf{G} = (G, (n+1, \ldots, n+k), (1, \ldots, k))$*, where* G *is a NUIG on* $[n+k]$ *whose first* k *and last* k *vertices form cliques.*

Proof. Using the arc representation of the proper circular arc graph, pick some point p on the circle that is not an endpoint of an arc, and intersects k intervals which we label $1, \ldots, k$ counterclockwise by their starting points. Then, cutting the circle at p and unraveling the arcs onto the number line produces exactly the NUIG we need (after extending the terminal intervals to ensure no interval contains another), shown in Figure 2. □

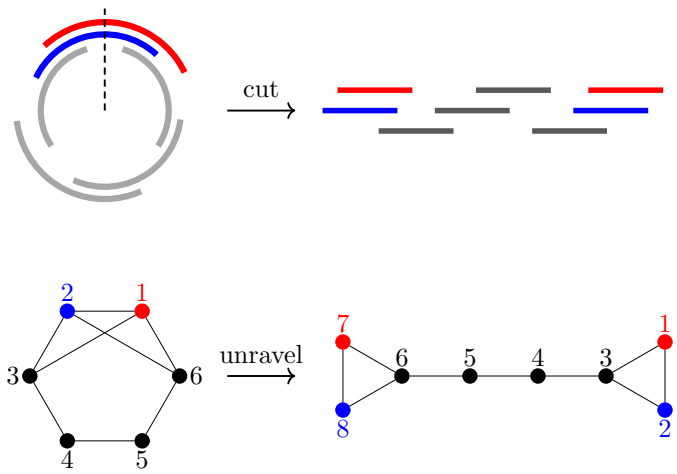


Figure 2: A proper circular arc graph being unraveled into a NUIG.

3 Color matrices

Definition 7. Let $I_k = \{(i_1, \ldots, i_k) \in [N]^k : i_1, \ldots, i_k \text{ distinct}\}$. For finite N, the spaces of *color vectors* $\hat{W}_k$ and $\hat{V}_k$ are spanned by the $\hat{b}_i$ with $i \in [N]^k$ and $i \in I_k$

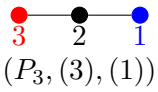
respectively. For $N = \infty$, $\hat{V}_k$ consists of the infinite sums $\sum_{i \in I_k} F(x_{i_1}, \dots, x_{i_k})\, \hat{b}_i$, where $F \in x_1 \cdots x_k\, \mathbb{Q}(e_1, e_2, \dots)[x_1, \dots, x_k]$. When $N = \infty$, the trace is the formal sum of the diagonal coefficients; our trace arguments use only finite N.

Definition 8. The *color matrix* of a marked graph $\mathbf{G} = (G, \alpha, \beta)$ is $\hat{M}_{\mathbf{G}} \colon \hat{W}_{\ell(\beta)} \to \hat{W}_{\ell(\alpha)}$,

$$\hat{M}_{\mathbf{G}} = \sum_{\kappa \text{ proper}} \frac{\prod_{v \in V(G)} x_{\kappa(v)}}{\prod_r x_{\kappa(\beta_r)}}\, \hat{b}_{\kappa(\alpha)}\, \hat{b}^T_{\kappa(\beta)},$$

where $\kappa(\alpha) = (\kappa(\alpha_1), \dots, \kappa(\alpha_\ell))$. If α and β are cliques, $\hat{M}_{\mathbf{G}}$ restricts to a map $\hat{V}_{\ell(\beta)} \to \hat{V}_{\ell(\alpha)}$, which extends to $N = \infty$.

Example 9. Figure 3 shows the colors corresponding to $\hat{M}_{(P_3,(3),(1))}$ for $N = 3$.



$(P_3, (3), (1))$

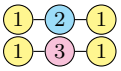
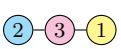
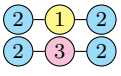
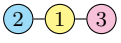
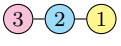
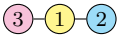
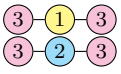
	$\kappa(1) = 1$	$\kappa(1) = 2$	$\kappa(1) = 3$
$\kappa(3) = 1$	 $x_1(x_2 + x_3)$	 x_1x_3	 x_1x_2
$\kappa(3) = 2$	 x_2x_3	 $x_2(x_1 + x_3)$	 x_2x_1
$\kappa(3) = 3$	 x_3x_2	 x_3x_1	 $x_3(x_1 + x_2)$

Figure 3: The color matrix of $(P_3, (3), (1))$ with $N = 3$.

Lemma 10. $\hat{M}_{\mathbf{G}+\mathbf{H}} = \hat{M}_{\mathbf{G}} \hat{M}_{\mathbf{H}}$. *If α and β are disjoint cliques of the same length, then* $\operatorname{tr} \hat{M}_{\mathbf{G}} = X_{\operatorname{tr} \mathbf{G}}(\boldsymbol{x})$, *the trace taken on* $\hat{V}_{\ell(\beta)}$.

Proof. This is [12, Lemma 3.21] with weights: since a proper coloring of $\mathbf{G} + \mathbf{H}$ is a pair of proper colorings of G and H agreeing on the glued vertices,

$$(\hat{M}_{\mathbf{G}+\mathbf{H}})_{i,j} = \sum_{\substack{\kappa(\alpha)=i \\ \kappa(\beta')=j}} \prod_{v \notin \beta'} x_{\kappa(v)} = \sum_l \Big(\sum_{\substack{\kappa(\alpha)=i \\ \kappa(\beta)=l}} \prod_{v \notin \beta} x_{\kappa(v)} \Big) \Big(\sum_{\substack{\kappa'(\alpha')=l \\ \kappa'(\beta')=j}} \prod_{v \notin \beta'} x_{\kappa'(v)} \Big) = (\hat{M}_{\mathbf{G}} \hat{M}_{\mathbf{H}})_{i,j},$$

$$\operatorname{tr} \hat{M}_{\mathbf{G}} = \sum_i \sum_{\kappa(\alpha)=\kappa(\beta)=i} \prod_{v \notin \beta} x_{\kappa(v)} = X_{\operatorname{tr} \mathbf{G}}(\boldsymbol{x}).$$

□

Definition 11. Let K_m be the complete graph on $[m]$, and let

$$\hat{M}_k = \hat{M}_{(K_{k+1},\, (1,\dots,k+1),\, (1,\dots,k))}, \qquad \hat{D}_k = \hat{M}_{(K_k,\, (2,\dots,k),\, (1,\dots,k))},$$

so that $\hat{M}_k \hat{b}_i = \sum_{c \notin i} x_c\, \hat{b}_{i,c}$ and $\hat{D}_k \hat{b}_i = \hat{b}_{i_2,\dots,i_k}$.

These two matrices build every graph of Lemma 6. More generally, if G is a NUIG on $[n]$ whose first a and last b vertices form cliques, we call $\mathbf{G} = (G, \alpha, \beta)$ with $\alpha = (n-b+1, \dots, n)$ and $\beta = (1, \dots, a)$ a *marked NUIG*.

Lemma 12. *If* $\mathbf{G}$ *is a marked NUIG, then* $\hat{M}_{\mathbf{G}} = \hat{X}_1 \cdots \hat{X}_m$*, where each* $\hat{X}_s$ *is some* $\hat{M}_j$ *or* $\hat{D}_j$*. Hence, if* $\mathbf{G}$ *is as in Lemma 6, Lemma 10 gives*

$$X_{\operatorname{tr}\mathbf{G}}(\boldsymbol{x}) = \operatorname{tr}(\hat{M}_{\mathbf{G}}). \tag{1}$$

Proof. We build G one vertex at a time, starting with the inputs $1, \dots, a$ marked. By the NUIG condition, the earlier neighbors of each new vertex are the r most recently marked vertices for some r, so we apply $\hat{D}$ to forget the older marked vertices and then $\hat{M}_r$ to attach the new vertex. Finally, we apply $\hat{D}$ until only the outputs $n-b+1, \dots, n$ remain. By Lemma 10, $\hat{M}_{\mathbf{G}}$ is the product of these steps, with later steps on the left. □

Example 13. In Figure 4, the inputs $1, 2$ are blue and the marked clique is red. The rightmost graph is built by $\hat{M}_3\hat{M}_2\hat{D}_3\hat{M}_2\hat{D}_3\hat{M}_2$. To add vertex 7, whose earlier neighbors are $4, 5, 6$, we first forget vertex 3 with $\hat{D}_4$ and then attach 7 with $\hat{M}_3$, so that $4, \dots, 7$ are marked.

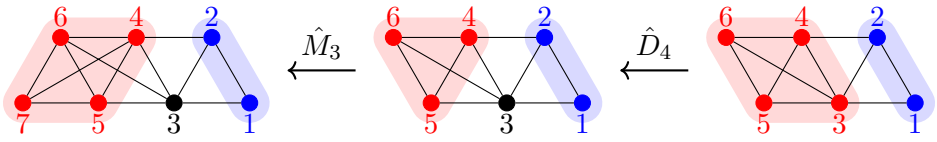


Figure 4: Adding vertex 7 with $\hat{M}_3\hat{D}_4$.

4 Tableau operators

Let V_k be the $\mathbb{Q}(e_1, e_2, \dots)$-vector space with basis $\{b_\sigma : \sigma \in I_k\}$. In Sections 4 and 5, $N = \infty$, so I_k is the set of k-tuples of distinct positive integers.

Definition 14. For $\sigma \in I_k$, let $T(\sigma) = \{y \notin \sigma : \ y = 1 \text{ or } y - 1 \in \sigma\}$, and for $y \in T(\sigma)$ let

$$A(1 \mid \sigma) = \prod_r \frac{\sigma_r - 1}{\sigma_r}, \qquad A(\sigma_r + 1 \mid \sigma) = \frac{\sigma_r + 1}{\sigma_r} \prod_{r' \neq r} \frac{\sigma_r - \sigma_{r'} + 1}{\sigma_r - \sigma_{r'}},$$

and $A(y \mid \sigma) = 0$ otherwise. The *tableau matrices* $M_k \colon V_k \to V_{k+1}$ and $D_k \colon V_k \to V_{k-1}$ are

$$M_k b_\sigma = \sum_y A(y \mid \sigma)\, b_{\sigma,y}, \qquad D_k b_\sigma = \frac{e_{\sigma_1}}{e_{\sigma_1 - 1}}\, b_{\sigma_2, \dots, \sigma_k}.$$

The entries of M_k are nonnegative rationals, and those of D_k are ratios e_a/e_{a-1}. If $a \notin \sigma$, then only the factor coming from a is new, so for every $y \neq a+1$,

$$A(y \mid a, \sigma) = \frac{y - a}{y - a - 1}\, A(y \mid \sigma). \tag{2}$$

Theorem 15. *There are linear maps $U_k\colon V_k \to \hat{V}_k$, $k \geq 0$, with $U_0 = 1$, such that*

$$U_k M_{k-1} = \hat{M}_{k-1} U_{k-1}, \tag{M}$$

$$\hat{D}_k U_k = U_{k-1} D_k. \tag{D}$$

Definition 16. Let $\mathbf{G}$ be a marked NUIG with $\hat{M}_{\mathbf{G}} = \hat{X}_1 \cdots \hat{X}_m$ as in Lemma 12. The *tableau matrix* of $\mathbf{G}$ is $M_{\mathbf{G}} = X_1 \cdots X_m$, where X_s is $\hat{X}_s$ with the hat removed.

Applying Theorem 15 to each factor gives $U_b M_{\mathbf{G}} = \hat{M}_{\mathbf{G}} U_a$. Section 7 describes $M_{\mathbf{G}}$ directly using Hikita's tableaux.

Example 17. Let $N = 3$ and $\mathbf{G} = (P_3, (3), (1))$. By Lemma 12, $\hat{M}_{\mathbf{G}} = (\hat{D}_2 \hat{M}_1)^2$ is the matrix in Figure 3. On the tableau side, with rows and columns indexed by b_1, b_2, b_3 (dropping b_4, as in Section 6),

$$D_2 M_1 = \begin{pmatrix} 0 & \frac{e_2}{2e_1} & \frac{2e_3}{3e_2} \\ 2e_1 & 0 & 0 \\ 0 & \frac{3e_2}{2e_1} & 0 \end{pmatrix}, \qquad (D_2 M_1)^2 = \begin{pmatrix} e_2 & \frac{e_3}{e_1} & 0 \\ 0 & e_2 & \frac{4e_1 e_3}{3e_2} \\ 3e_2 & 0 & 0 \end{pmatrix}.$$

The change of basis from Section 5 is

$$U_1 = \begin{pmatrix} x_1 & \frac{x_1(x_2+x_3)}{2e_1} & \frac{e_3}{3e_2} \\ x_2 & \frac{x_2(x_1+x_3)}{2e_1} & \frac{e_3}{3e_2} \\ x_3 & \frac{x_3(x_1+x_2)}{2e_1} & \frac{e_3}{3e_2} \end{pmatrix},$$

and $U_1 (D_2 M_1)^2 = \hat{M}_{\mathbf{G}} U_1$. As a check,

$$\begin{aligned} D_1 (D_2 M_1)^2 M_0 &= \begin{pmatrix} e_1 & \frac{e_2}{e_1} & \frac{e_3}{e_2} \end{pmatrix} \begin{pmatrix} e_2 & \frac{e_3}{e_1} & 0 \\ 0 & e_2 & \frac{4e_1 e_3}{3e_2} \\ 3e_2 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \\ &= e_1 e_2 + 3e_3 = X_{P_3}(\boldsymbol{x}) = \hat{D}_1 \hat{M}_{\mathbf{G}} \hat{M}_0, \end{aligned}$$

and both traces equal $2e_2 = X_{\operatorname{tr} \mathbf{G}}(\boldsymbol{x})$, where $\operatorname{tr} \mathbf{G}$ is P_2.

5 Proof of the main theorem

We will use induction. To construct the inductive hypothesis, we define

$$R_k = M_{k-1} D_k - D_{k+1} M_k, \qquad \hat{R}_k = \hat{M}_{k-1} \hat{D}_k - \hat{D}_{k+1} \hat{M}_k,$$

and, for $1 \leq s < k$,

$$S_s b_\sigma = \frac{1}{\sigma_{s+1} - \sigma_s} b_\sigma + \left(1 - \frac{1}{\sigma_{s+1} - \sigma_s}\right) b_{\tau_s \sigma}, \qquad \hat{S}_s \hat{b}_i = \hat{b}_{\tau_s i},$$

where $\tau_s(\sigma_1,\dots,\sigma_k) = (\sigma_1,\dots,\sigma_{s-1},\sigma_{s+1},\sigma_s,\sigma_{s+2},\dots,\sigma_k)$ swaps the entries in positions s and $s+1$. In particular, $\hat{S}$ applies an adjacent swap and $\hat{R}_k\hat{b}_i = x_{i_1}\hat{b}_{i_2,\dots,i_k,i_1}$ applies a cyclic permutation. To simplify R_k, let $a \in \mathbb{N}$ and $w \in I_{k-1}$, with $a \notin w$, and let (a,w) denote the tuple obtained by prepending a to w. Direct substitution in the definition of R_k yields

$$\begin{aligned} R_k b_{a,w} &= \frac{e_a}{e_{a-1}} \sum_y \big(A(y \mid w) - A(y \mid a,w)\big) b_{w,y} \\ &= \frac{e_a}{e_{a-1}} \left(\left(\sum_{y \in T(w)} \frac{A(y \mid w)}{a+1-y} b_{w,y} \right) - A(a{+}1 \mid a,w) b_{w,a+1} \right). \end{aligned} \tag{3}$$

We use the following identities for induction.

Proposition 18. *For $k \geq 2$, the following hold for $\{M, D, R, S\}$ and $\{\hat{M}, \hat{D}, \hat{R}, \hat{S}\}$:*

$$S_s M_{k-1} = M_{k-1} S_s, \qquad S_s R_k = R_k S_{s+1} \qquad (1 \leq s \leq k-2), \tag{4}$$

$$D_k R_k = R_{k-1} D_k S_1, \tag{5}$$

$$S_{k-1} M_{k-1} R_{k-1} = R_k M_{k-1}, \qquad S_{k-1} M_{k-1} M_{k-2} = M_{k-1} M_{k-2}, \tag{6}$$

$$S_{k-1} R_k^2 = R_k^2 S_1. \tag{7}$$

Proof. Equation (4) follows because $A(y \mid \sigma)$ depends only on the set of entries of σ. For Equation (5), let $\sigma \in I_{k-2}$ and let $a, b \notin \sigma$ be distinct, then

$$\begin{aligned} D_k R_k b_{a,b,\sigma} &= \frac{e_a e_b}{e_{a-1} e_{b-1}} \sum_{y \notin \sigma} \big(A(y \mid b,\sigma) - A(y \mid a,b,\sigma)\big) b_{\sigma,y} \\ &= \frac{e_a e_b}{e_{a-1} e_{b-1}} \sum_{y \notin \sigma} \Bigg(\frac{A(y \mid \sigma) - A(y \mid b,\sigma)}{b-a} \\ &\qquad + \left(1 - \frac{1}{b-a}\right) \big(A(y \mid \sigma) - A(y \mid a,\sigma)\big) \Bigg) b_{\sigma,y} \\ &= \frac{1}{b-a} \frac{e_a}{e_{a-1}} R_{k-1} b_{b,\sigma} + \left(1 - \frac{1}{b-a}\right) \frac{e_b}{e_{b-1}} R_{k-1} b_{a,\sigma} \\ &= R_{k-1} D_k S_1 b_{a,b,\sigma}. \end{aligned}$$

For Equation (6), let $\sigma \in I_{k-2}$, let y, z range over distinct elements of $T(\sigma)$, and let $P = \sum_y A(y \mid \sigma) A(y{+}1 \mid \sigma, y)\, b_{\sigma,y,y+1}$, which S_{k-1} fixes, then

$$\begin{aligned} S_{k-1} M_{k-1} M_{k-2} b_\sigma &= S_{k-1} \Big(P + \sum_{y,z} \frac{z-y}{z-y-1} A(y \mid \sigma) A(z \mid \sigma)\, b_{\sigma,y,z} \Big) \\ &= P + \sum_{y,z} A(y \mid \sigma) A(z \mid \sigma) \Big(\frac{b_{\sigma,y,z}}{z-y-1} + b_{\sigma,z,y} \Big) \\ &= P + \sum_{y,z} \frac{z-y}{z-y-1} A(y \mid \sigma) A(z \mid \sigma)\, b_{\sigma,y,z} = M_{k-1} M_{k-2} b_\sigma. \end{aligned}$$

For its first identity, expanding the products gives

$$\begin{aligned} R_k M_{k-1} b_{a,\sigma} &= \frac{e_a}{e_{a-1}} \sum_{\substack{y,z \notin \sigma \\ y \neq z,\ y \neq a}} A(y \mid a, \sigma)\big(A(z \mid \sigma, y) - A(z \mid a, \sigma, y)\big) b_{\sigma,y,z} \\ &= S_{k-1} M_{k-1} R_{k-1} b_{a,\sigma}. \end{aligned}$$

For the second equality, if $y \neq a+1$ and $z \notin \{y+1, a+1\}$, Equation (2) gives both sides the coefficient $-\frac{e_a}{e_{a-1}} A(y \mid \sigma) A(z \mid \sigma) \frac{(y-a)(z-y)}{(y-a-1)(z-y-1)(z-a-1)}$ on $b_{\sigma,y,z}$. The remaining cases are similar. The hatted identities follow directly by appending, deleting, and swapping colors; for example,

$$\begin{aligned} \hat{D}_k \hat{R}_k \hat{b}_{a,b,\sigma} &= x_a \hat{b}_{\sigma,a} = \hat{R}_{k-1} \hat{D}_k \hat{S}_1 \hat{b}_{a,b,\sigma}, \\ \hat{S}_{k-1} \hat{M}_{k-1} \hat{R}_{k-1} \hat{b}_{a,\sigma} &= x_a \sum_{c \notin (a,\sigma)} x_c \hat{b}_{\sigma,c,a} = \hat{R}_k \hat{M}_{k-1} \hat{b}_{a,\sigma}. \end{aligned}$$

Finally, $S_s^2 = 1$ and $S_s D_{k+1} = D_{k+1} S_{s+1}$ directly from the definitions. Equations (6), (5), and (4) now give (for the hatted identities as well)

$$\begin{aligned} S_{k-1} R_k^2 &= S_{k-1} R_k M_{k-1} D_k - S_{k-1} R_k D_{k+1} M_k \\ &= M_{k-1} R_{k-1} D_k - D_{k+1} M_k R_k S_1 \\ &= M_{k-1} D_k R_k S_1 - D_{k+1} M_k R_k S_1 \\ &= R_k^2 S_1, \end{aligned}$$

proving Equation (7). □

Proof of Theorem 15. We construct U_k satisfying Equations (M) and (D), together with

$$U_k R_k = \hat{R}_k U_k, \tag{R}$$

$$U_k S_s = \hat{S}_s U_k \quad (1 \leq s < k). \tag{S}$$

For $k = 1$, let $e_r(\boldsymbol{x} \setminus x_c)$ denote the elementary symmetric function of degree r in the variables $(x_d)_{d \neq c}$, and define

$$U_1 b_a = \frac{1}{a\, e_{a-1}} \sum_c x_c\, e_{a-1}(\boldsymbol{x} \setminus x_c)\, \hat{b}_c.$$

With rows indexed by the $\hat{b}_c$ and columns by the b_a, its upper-left corner is

$$U_1 = \begin{pmatrix} x_1 & \dfrac{x_1(e_1 - x_1)}{2e_1} & \cdots \\ x_2 & \dfrac{x_2(e_1 - x_2)}{2e_1} & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}.$$

The definitions give

$$R_1 b_a = \frac{e_a}{e_{a-1}} \left(\frac{1}{a} b_1 - \frac{a+1}{a} b_{a+1} \right), \qquad \hat{R}_1 \hat{b}_c = x_c \hat{b}_c.$$

Together with

$$e_a - e_a(\boldsymbol{x} \setminus x_c) = x_c e_{a-1}(\boldsymbol{x} \setminus x_c), \qquad \sum_c x_c e_{a-1}(\boldsymbol{x} \setminus x_c) = a e_a,$$

this yields

$$\begin{aligned}
U_1 M_0 b_\emptyset = U_1 b_1 &= \sum_c x_c \hat{b}_c = \hat{M}_0 U_0 b_\emptyset, \\
\hat{D}_1 U_1 b_a &= \frac{1}{a e_{a-1}} \sum_c x_c e_{a-1}(\boldsymbol{x} \setminus x_c)\, \hat{b}_\emptyset = \frac{e_a}{e_{a-1}} \hat{b}_\emptyset = U_0 D_1 b_a, \\
U_1 R_1 b_a &= \frac{1}{a e_{a-1}} \sum_c x_c^2 e_{a-1}(\boldsymbol{x} \setminus x_c)\, \hat{b}_c = \hat{R}_1 U_1 b_a.
\end{aligned}$$

Thus Equations (M), (D), and (R) hold, while Equation (S) is vacuous. The formula for $R_1 b_a$, together with $b_1 = M_0 b_\emptyset$, also gives $V_1 = M_0 V_0 + R_1 V_1$, the case $k = 1$ of Equation (11) below.

Now fix $k \geq 2$ and assume the claims below level k. We construct U_k by induction on its columns. Equation (R) applied to $b_{a,w}$, expanded using Equation (3), involves $U_k b_{a,w}$, the columns $U_k b_{w,y}$ with $y \in T(w)$, and, when $a+1 \notin w$, the column $U_k b_{w,a+1}$; the $T(w)$-columns occur together in Equation (M). Thus, by ordering the basis vectors indexed by I_k, we can use the two equations inductively to construct U_k. For $w \in I_{k-1}$ and $y \notin w$, give $b_{w,y}$ the rank $r(w,y) = (r_j(w,y))_{j \geq 1}$, where

$$r_j(w,y) = \begin{cases} k+1-s, & j = w_s, \\ 1, & j = y,\ y \notin T(w), \\ 0, & \text{otherwise.} \end{cases}$$

We compare $r(w,y) < r(w',y')$ if $r_j(w,y) < r_j(w',y')$ where j is the largest differing index. Roughly, $r(w,y) < r(w',y')$ if w' has larger entries than w, and moving large entries to the right lowers the rank. Each rank has finitely many predecessors, so we can fix $w \in I_{k-1}$ and assume inductively that U_k is defined on every column of smaller rank. The columns $U_k b_{w,y}$ for $y \in T(w)$ have the same rank, and we solve for them simultaneously. Let $Z(w) = \{z \in w : z \geq 2,\ z-1 \notin w\}$. Equation (M) applied to b_w and Equation (R) applied to $b_{z-1,w}$ for $z \in Z(w)$ give

$$\sum_{y \in T(w)} A(y \mid w) U_k b_{w,y} = \hat{M}_{k-1} U_{k-1} b_w, \tag{8}$$

$$\sum_{y \in T(w)} \frac{A(y \mid w)}{z-y} U_k b_{w,y} = \frac{e_{z-2}}{e_{z-1}} \hat{R}_k U_k b_{z-1,w} \quad (\forall z \in Z(w)). \tag{9}$$

The column $b_{z-1,w}$ has smaller rank than $b_{w,y}$ for $y \in T(w)$. The right side of Equation (8) is known from U_{k-1}, and those of Equation (9) from the induction on rank. Unpacking T and Z gives $|T(w)| = |Z(w)| + 1$, so the system is square. To show the coefficient matrix is invertible, note $A(y \mid w) \neq 0$, so a null vector $(c_y)_{y \in T(w)}$ must satisfy

$$\sum_{y \in T(w)} c_y = 0, \qquad \sum_{y \in T(w)} \frac{c_y}{z - y} = 0 \quad (\forall z \in Z(w)).$$

For $z \in Z(w)$, note

$$\sum_{y \in T(w)} \frac{c_y}{z - y} = \frac{P(z) + \sum_{y \in T(w)} c_y z^{|T(w)|-1}}{Q(z)} = \frac{P(z)}{Q(z)} = 0,$$

where P has degree at most $|T(w)| - 2$ and $Q(z) = \prod_{y \in T(w)} (z - y) \neq 0$ since $Z(w) \cap T(w) = \emptyset$. Thus $P(z) = 0$ for all $z \in Z(w)$, so $P = 0$ by its degree bound. Plugging in $z = y \in T(w)$

$$0 = P(y) = \sum_{y' \in T(w)} c_{y'} \prod_{y'' \in T(w) \setminus \{y'\}} (y - y'') = c_y \prod_{y' \in T(w) \setminus \{y\}} (y - y'),$$

forcing $c_y = 0$, meaning the only null vector is 0. Thus the system is invertible and determines $U_k b_{w,y}$ for $y \in T(w)$.

For the remaining $b_{w,y}$ with $y \notin T(w)$ and $y \notin w$, Equation (R) gives

$$A(y \mid y-1, w) U_k b_{w,y} = \left(\sum_{z \in T(w)} \frac{A(z \mid w)}{y - z} U_k b_{w,z} \right) - \frac{e_{y-2}}{e_{y-1}} \hat{R}_k U_k b_{y-1,w}. \tag{10}$$

The columns $U_k b_{w,z}$ were determined above, and $r(y-1, w) < r(w, y)$: the largest entry of (w, y) moves right if it lies in w, and disappears otherwise.

Thus $U_k b_{y-1,w}$ is known by induction. Since $A(y \mid y-1, w) \neq 0$, this equation determines the remaining columns. Every instance of Equations (M) and (R) is used exactly once, so both hold. Inverting the coefficient matrix in Equations (8) and (9) also expresses $b_{w,y}$, $y \in T(w)$, as linear combinations of $M_{k-1} b_w$ and $R_k b_{z-1,w}$, $z \in Z(w)$. Equation (3) expresses each remaining $b_{w,y}$ using $b_{w,z}$, $z \in T(w)$, and $R_k b_{y-1,w}$, whose source columns have smaller rank. Thus, for every $r(w, y) = r$,

$$\begin{aligned} b_{w,y} &\in \operatorname{span}\big(\{M_{k-1} b_w\} \cup \{R_k b_{a,w} : a \in \mathbb{N} \setminus w,\ a+1 \in w \cup \{y\}\}\big) \\ &\subseteq M_{k-1} V_{k-1} + \operatorname{span}\{R_k b_\sigma : r(\sigma) < r\}. \end{aligned} \tag{11}$$

With U_k constructed, we prove Equation (S) by induction on rank, keeping k fixed. The inductive hypothesis at rank r states for every b_σ with $r(\sigma) < r$,

$$\begin{aligned} U_k S_s b_\sigma &= \hat{S}_s U_k b_\sigma \qquad (1 \leq s < k), \\ U_k S_{k-1} R_k b_\sigma &= \hat{S}_{k-1} \hat{R}_k U_k b_\sigma. \end{aligned} \tag{12}$$

To establish the two equalities at rank r, Equation (11) reduces the check to

$$\begin{aligned} U_k S_s M_{k-1} &= \hat{S}_s U_k M_{k-1}, & U_k S_s R_k b_\sigma &= \hat{S}_s U_k R_k b_\sigma & (\forall r(\sigma) < r),\\ U_k S_{k-1} R_k M_{k-1} &= \hat{S}_{k-1}\hat{R}_k U_k M_{k-1}, & U_k S_{k-1} R_k^2 b_\sigma &= \hat{S}_{k-1}\hat{R}_k U_k R_k b_\sigma & (\forall r(\sigma) < r). \end{aligned}$$

On $M_{k-1}V_{k-1}$, Equation (4) and induction on k give, for $s \le k-2$,

$$U_k S_s M_{k-1} = \hat{S}_s U_k M_{k-1}.$$

For $s = k-1$, Equation (6) and the established Equations (M) and (R) give

$$\begin{aligned} U_k S_{k-1} M_{k-1} M_{k-2} &= \hat{S}_{k-1} U_k M_{k-1} M_{k-2},\\ U_k S_{k-1} M_{k-1} R_{k-1} &= \hat{S}_{k-1} U_k M_{k-1} R_{k-1}, \end{aligned}$$

which combined with Equation (11) at level $k-1$ gives $U_k S_{k-1} M_{k-1} = \hat{S}_{k-1} U_k M_{k-1}$. The second required equality follows from Equation (6) and $S_{k-1}^2 = \hat{S}_{k-1}^2 = 1$:

$$U_k S_{k-1} R_k M_{k-1} = U_k M_{k-1} R_{k-1} = \hat{M}_{k-1}\hat{R}_{k-1} U_{k-1} = \hat{S}_{k-1}\hat{R}_k U_k M_{k-1}.$$

On $R_k b_\sigma$ with $r(\sigma) < r$, the inductive hypothesis and Equations (4) and (7) give the remaining required equalities:

$$\begin{aligned} U_k S_s R_k b_\sigma &= \hat{R}_k U_k S_{s+1} b_\sigma = \hat{S}_s U_k R_k b_\sigma & (1 \le s \le k-2),\\ U_k S_{k-1} R_k b_\sigma &= \hat{S}_{k-1}\hat{R}_k U_k b_\sigma = \hat{S}_{k-1} U_k R_k b_\sigma,\\ U_k S_{k-1} R_k^2 b_\sigma &= \hat{R}_k^2 U_k S_1 b_\sigma = \hat{S}_{k-1}\hat{R}_k U_k R_k b_\sigma, \end{aligned}$$

proving Equation (S). Finally, we prove Equation (D) by a similar induction on rank, keeping k fixed. Fix a rank r. The inductive hypothesis is

$$\hat{D}_k U_k b_\sigma = U_{k-1} D_k b_\sigma \qquad (\forall r(\sigma) < r).$$

Yet again, to prove this equality for $b_{w,y}$ of rank r, Equation (11) reduces this to showing

$$\begin{aligned} \hat{D}_k U_k M_{k-1} &= U_{k-1} D_k M_{k-1},\\ \hat{D}_k U_k R_k b_{a,w} &= U_{k-1} D_k R_k b_{a,w} \qquad (\forall a \in \mathbb{N} \setminus w,\ a+1 \in w \cup \{y\}). \end{aligned}$$

On $M_{k-1}V_{k-1}$, the definition of R_{k-1} and induction on k give

$$\begin{aligned} \hat{D}_k U_k M_{k-1} &= (\hat{M}_{k-2}\hat{D}_{k-1} - \hat{R}_{k-1}) U_{k-1}\\ &= U_{k-1}(M_{k-2} D_{k-1} - R_{k-1}) = U_{k-1} D_k M_{k-1}. \end{aligned}$$

On $R_k b_{a,w}$, Equations (5), (R), and (S) give

$$\begin{aligned} \hat{D}_k U_k R_k b_{a,w} &= \hat{R}_{k-1}\hat{D}_k U_k S_1 b_{a,w},\\ U_{k-1} D_k R_k b_{a,w} &= \hat{R}_{k-1} U_{k-1} D_k S_1 b_{a,w}. \end{aligned}$$

The vector $S_1 b_{a,w}$ only involves basis vectors of rank smaller than r: if $a+1 = w_1$, then $S_1 b_{a,w} = b_{a,w}$, and otherwise $\tau_1(a,w) = (w_1, a, w_2, \ldots, w_{k-1})$ has smaller rank, since the entries after w_1 move right, y drops out, and $a < a+1 \in w \cup \{y\}$. Hence the inductive hypothesis makes the two displayed right sides equal. Therefore

$$\hat{D}_k U_k R_k b_{a,w} = U_{k-1} D_k R_k b_{a,w} \qquad (\forall a \in \mathbb{N} \setminus w,\ a+1 \in w \cup \{y\}).$$

This completes the induction, proving Equation (D) and Theorem 15. □

6 Traces and e-positivity

We have found a change of basis family U_k that relates the color matrices to tableau matrices. To transport the trace property from coloring to tableau, we need to show that when restricted to a finite yet sufficient amount of colors, the maps in Theorem 15 are invertible.

We fix $N \in \mathbb{N}$ and work over $\mathbb{Q}(x_1, \ldots, x_N)$, so I_k is now the set of k-tuples of distinct entries in $[N]$, and V_k has basis $\{b_\sigma : \sigma \in I_k\}$. Let $M_k^{(N)}$ be the restriction of M_k to these tuples, omitting any term $b_{\sigma,N+1}$, and let D_k act by the same formula as before, with $e_j = e_j(x_1, \ldots, x_N)$. We want to find a map $U_k^{(N)}$ that is invertible with

$$U_k^{(N)} M_{k-1}^{(N)} = \hat{M}_{k-1} U_{k-1}^{(N)}, \qquad \hat{D}_k U_k^{(N)} = U_{k-1}^{(N)} D_k.$$

It turns out $U_k^{(N)}$ comes from setting $x_c = 0$ for $c > N$ in U_k. We need to show this is well-defined, that its image lies in $\hat{V}_k$, and that it satisfies the two equations above.

First, the restriction is well-defined. Let σ be a k-tuple of distinct positive integers. By the definition of $\hat{V}_k$ with $N = \infty$, $U_k b_\sigma$ decomposes as

$$U_k b_\sigma = \sum_i F_\sigma(x_{i_1}, \ldots, x_{i_k}) \hat{b}_i,$$

where we will show $F_\sigma(x_{i_1}, \ldots, x_{i_k}) \in x_{i_1} \cdots x_{i_k}\, \mathbb{Q}[\boldsymbol{x}][e_j^{-1} : j < \max \sigma]$. This holds for U_1 and is preserved by $\hat{M}_{k-1}$ and $\hat{R}_k$, which multiply each new color by its variable. By induction on rank, Equations (8) and (9) divide only by e_j with $j < \max w$, and Equation (10) only by e_j with $j < \max(w, y)$.

Thus if σ has entries at most $N+1$, no e_j with $j > N$ appears in a denominator, and we may set $x_c = 0$ for $c > N$. This defines $U_k^{(N)} b_\sigma$ for every σ with entries at most $N+1$. In particular, $U_k^{(N)}$ is defined on V_k.

Second, the image lies in $\hat{V}_k$, since $x_{i_1} \cdots x_{i_k}$ divides each $F_\sigma(x_{i_1}, \ldots, x_{i_k})$, so every term with a color outside $[N]$ vanishes.

Third, specializing Equation (D) gives $\hat{D}_k U_k^{(N)} = U_{k-1}^{(N)} D_k$. Specializing Equation (M) at b_w with $w \in I_{k-1}$ gives

$$U_k^{(N)} M_{k-1} b_w = \hat{M}_{k-1} U_{k-1}^{(N)} b_w, \qquad M_{k-1} b_w = M_{k-1}^{(N)} b_w + A(N{+}1 \mid w)\, b_{w,N+1}.$$

We use the following lemma to show $U_k^{(N)} b_{w,N+1} = 0$, so $U_k^{(N)} M_{k-1}^{(N)} = \hat{M}_{k-1} U_{k-1}^{(N)}$.

Lemma 19. *If σ has entries at most $N+1$ and contains $N+1$, then $U_k^{(N)} b_\sigma = 0$.*

Proof. We use downward induction on k. For $k = N+1$, there are no color tuples with distinct entries, so $\hat{V}_{N+1} = 0$. Now suppose $k \le N$ and the claim holds at level $k+1$. Since σ contains $N+1$, there is an $a \in [N] \setminus \sigma$. Equation (D) gives

$$0 = \hat{D}_{k+1} U_{k+1}^{(N)} b_{a,\sigma} = \frac{e_a}{e_{a-1}} U_k^{(N)} b_\sigma,$$

proving the claim. □

From now on, $U_k^{(N)}$ denotes its restriction to V_k, so $U_k^{(N)} \colon V_k \to \hat{V}_k$ satisfies the two equations above.

Proposition 20. *For $0 \le k \le N$, the map $U_k^{(N)} \colon V_k \to \hat{V}_k$ is invertible. For $1 \le k \le N$,*

$$U_k^{(N)} M_{k-1}^{(N)} = \hat{M}_{k-1} U_{k-1}^{(N)}, \qquad \hat{D}_k U_k^{(N)} = U_{k-1}^{(N)} D_k.$$

Proof. Equations (M) and (D), together with Lemma 19, give the two displayed equalities. Next, since $\hat{R}_k = \hat{M}_{k-1}\hat{D}_k - \hat{D}_{k+1}\hat{M}_k$, note

$$\hat{R}_k U_k^{(N)} = U_k^{(N)} (M_{k-1}^{(N)} D_k - D_{k+1} M_k^{(N)}),$$

where $M_N^{(N)} = \hat{M}_N = 0$. This means $\hat{R}_k$ preserves the image of $U_k^{(N)}$.

We prove invertibility by showing $U_k^{(N)}$ is surjective by induction on k, starting with $U_0^{(N)} = 1$. The inductive hypothesis then states that $U_{k-1}^{(N)}$ is surjective, so for every $i \in I_{k-1}$ we can find a v such that $\hat{b}_i = U_{k-1}^{(N)} v$. Then $\hat{M}_{k-1}\hat{b}_i = U_k^{(N)} M_{k-1}^{(N)} v$ lies in the image of $U_k^{(N)}$, and since applying $\hat{R}_k$ preserves the image, so does

$$(\hat{R}_k^k)^u \hat{M}_{k-1} \hat{b}_i = (x_{i_1} \cdots x_{i_{k-1}})^u \sum_{c \in [N] \setminus i} x_c^{u+1} \hat{b}_{i,c} \qquad (0 \le u \le N-k).$$

For fixed i, the coefficient matrix is a Vandermonde matrix in the $N-k+1$ distinct variables x_c, up to nonzero row and column factors. These vectors therefore span every $\hat{b}_{i,c}$. This proves surjectivity, and both spaces have dimension $N!/(N-k)!$, so $U_k^{(N)}$ is invertible.

□

For a marked NUIG $\mathbf{G}$, let $M_{\mathbf{G}}^{(N)}$ be $M_{\mathbf{G}}$ with each M_j replaced by $M_j^{(N)}$.

Theorem 21. *Let $\Gamma = \operatorname{tr} \mathbf{G}$ be a proper circular arc graph on n vertices, where $\mathbf{G} = (G, (n+1, \ldots, n+k), (1, \ldots, k))$ as in Lemma 6. For any $N \in \mathbb{N}$,*

$$X_\Gamma(x_1, \ldots, x_N) = \operatorname{tr} M_{\mathbf{G}}^{(N)},$$

which is e-positive. In particular, Γ is e-positive.

Proof. If every level in the product defining $M_{\mathbf{G}}^{(N)}$ is at most N, then Proposition 20 applies to each factor, and with Lemma 12, which holds for every N,

$$\operatorname{tr} M_{\mathbf{G}}^{(N)} = \operatorname{tr}\big((U_k^{(N)})^{-1}\hat{M}_{\mathbf{G}}U_k^{(N)}\big) = \operatorname{tr}\hat{M}_{\mathbf{G}} = X_\Gamma(x_1,\dots,x_N).$$

Otherwise, the product passes through V_j and $\hat{V}_j$ for some $j > N$, which are both zero since $[N]$ has no j distinct colors, so $\operatorname{tr} M_{\mathbf{G}}^{(N)} = 0 = \operatorname{tr}\hat{M}_{\mathbf{G}} = X_\Gamma(x_1,\dots,x_N)$.

The entries of $M_j^{(N)}$ and D_j are Laurent polynomials in $e_1,\dots,e_N$ with nonnegative coefficients, so the trace is too. Since $e_1,\dots,e_N$ are algebraically independent and the trace is the polynomial $X_\Gamma(x_1,\dots,x_N)$, it is e-positive. Taking $N \geq n$ shows $X_\Gamma(\boldsymbol{x})$ is e-positive. □

Example 22. The proper circular arc graph Γ in Figure 2 is $\operatorname{tr}\mathbf{G}$ for $\mathbf{G} = (G,(7,8),(1,2))$, where G is the NUIG on [8] shown there. Building G as in Lemma 12 gives $\hat{M}_{\mathbf{G}} = \hat{D}_3\hat{M}_2(\hat{M}_1\hat{D}_2)^4\hat{D}_3\hat{M}_2$, so by Theorem 21 and cyclicity of the trace,

$$X_\Gamma(x_1,\dots,x_N) = \operatorname{tr}\big((M_1^{(N)}D_2)^4(D_3M_2^{(N)})^2\big) = 16e_1e_5 + 20e_2e_4 + 12e_3^2 + 84e_6.$$

When $k = 0$, we do not need Proposition 20, since $U_0 = 1$.

Theorem 23. *If G is a NUIG, then*

$$X_G(\boldsymbol{x}) = M_{(G,\emptyset,\emptyset)},$$

which is e-positive. This proves the e-positivity of unit interval graphs without the modular law characterization of [1] used in [10], and together with [9, Theorem 5.1] gives an alternate proof of the Stanley–Stembridge conjecture.

Proof. Since $U_0 = 1$,

$$X_G(\boldsymbol{x}) = \hat{M}_{(G,\emptyset,\emptyset)}U_0 = U_0M_{(G,\emptyset,\emptyset)} = M_{(G,\emptyset,\emptyset)}.$$

As in the proof of Theorem 21, this is a Laurent polynomial in the algebraically independent $e_1, e_2, \dots$ with nonnegative coefficients that is also a polynomial, so it is e-positive. □

Example 24. For the NUIG $G+H$ in Figure 1, building it as in Lemma 12 and applying Theorem 23 gives

$$\begin{aligned} X_{G+H}(\boldsymbol{x}) &= D_1D_2D_3M_2D_3M_2M_1D_2M_1D_2D_3M_2M_1M_0 \\ &= 8e_1e_3^2 + 88e_3e_4 + 24e_2e_5 + 40e_1e_6 + 56e_7, \end{aligned}$$

which is e-positive since every factor has entries that are Laurent polynomials in the e_i with nonnegative coefficients.

7 Tableau matrices of graphs

We now explain M_k and D_k using Hikita's tableaux, which gives a tableau matrix $M_{\mathbf{G}}$ for every marked NUIG. This generalizes [21, Section 4] to several input and output boxes, with a different normalization. Throughout, $N = \infty$.

We draw tableaux with rows increasing to the right and columns increasing upward.

Definition 25 ([10, Definitions 1.2 and 1.5])**.** For a tableau T, let $c_T(v)$ be the column of the box v, and for a sequence of vertices α, let $c_T(\alpha) = (c_T(\alpha_1), \ldots, c_T(\alpha_\ell))$. Let $N(v)$ be the sequence of smaller neighbors of v in increasing order. *Hikita's rule* at $q = 1$ adds v to the top of column y of T with probability $A(y \mid c_T(N(v)))/y$, and we write $T \leftarrow_y v$ for the result. For a NUIG G, adding the remaining vertices of G in increasing order builds T' from T with probability

$$\Pr(T' \mid T, G) = \prod_{v \in T' \setminus T} \frac{A\big(c_{T'}(v) \mid c_{T'}(N(v))\big)}{c_{T'}(v)}.$$

Let $\Pr(T' \mid G) = \Pr(T' \mid \emptyset, G)$.

This is Hikita's transition probability φ_y at $q = 1$ [10, Definitions 1.2 and 1.5], where his red columns are the columns ≤ 0 and those containing a neighbor of v: telescoping the factors of A over each run of consecutive entries of $c_T(N(v))$ gives his formula. Since the smaller neighbors of v are at the tops of distinct columns, v can be added on top of column y whenever $A(y \mid c_T(N(v))) \neq 0$.

Theorem 26 ([10, Theorem 1.6] at $q = 1$)**.** *For a NUIG G,*

$$X_G(\boldsymbol{x}) = \sum_T \Pr(T \mid G) \prod_r \lambda_r!\, e_{\lambda_r},$$

where T has shape λ.

We combine the probabilities and the elementary symmetric functions into a single weight.

Definition 27. For a NUIG G and T, T' as in Definition 25, the *weight* of T' is

$$w(T' \mid T, G) = \prod_{v \in T' \setminus T} A\big(c_{T'}(v) \mid c_{T'}(N(v))\big) \frac{e_{c_{T'}(v)}}{e_{c_{T'}(v)-1}}.$$

Adding a box in column y turns a row of length $y - 1$ into one of length y, so the factors ye_y/e_{y-1} along each row telescope, and

$$w(T' \mid T, G) = \Pr(T' \mid T, G) \frac{\prod_r \lambda'_r!\, e_{\lambda'_r}}{\prod_r \lambda_r!\, e_{\lambda_r}}, \tag{13}$$

where λ and λ' are the shapes of T and T'.

Definition 28. Let $\mathbf{G} = (G, \alpha, \beta)$ be a marked NUIG, with $\beta = (1, \dots, a)$ and $\alpha = (n-b+1, \dots, n)$. For $j \in I_a$, let T_j be a tableau with rows of lengths $j_1, \dots, j_a$ and $c_{T_j}(\beta) = j$ (whose other entries are distinct and nonpositive), where $T_\emptyset = \emptyset$. With $\varepsilon(\sigma) = \prod_r e_{\sigma_r}/e_{\sigma_r - 1}$, let $\mathcal{M}_{\mathbf{G}} \colon V_a \to V_b$ be

$$\mathcal{M}_{\mathbf{G}} b_j = \sum_T w(T \mid T_j, G)\, \frac{\varepsilon(j)}{\varepsilon(c_T(\alpha))}\, b_{c_T(\alpha)},$$

where T ranges over tableaux containing T_j whose other entries are $a+1, \dots, n$, with $c_T(\alpha) \in I_b$. Since the weight only depends on the columns of the vertices of G, the choice of nonpositive entries does not matter.

Lemma 29. *If $\mathbf{G} = (G, \alpha, \beta)$ and $\mathbf{H} = (H, \alpha', \beta')$ are marked NUIGs and β, α' have the same length, then $\mathcal{M}_{\mathbf{G}+\mathbf{H}} = \mathcal{M}_{\mathbf{G}} \mathcal{M}_{\mathbf{H}}$.*

Proof. Building $G + H$ from T_j first adds the vertices of H, giving some T, and then those of G, giving some T'. Each $N(v)$ is the same in $G + H$ as in G, so $w(T' \mid T_j, G+H) = w(T \mid T_j, H)\, w(T' \mid T, G)$, and the second factor is the same as building G from T_i, where $i = c_T(\alpha')$. Grouping by i,

$$\mathcal{M}_{\mathbf{G}+\mathbf{H}} b_j = \sum_i \Big(\sum_{c_T(\alpha') = i} w(T \mid T_j, H)\, \frac{\varepsilon(j)}{\varepsilon(i)} \Big) \mathcal{M}_{\mathbf{G}} b_i = \mathcal{M}_{\mathbf{G}} \mathcal{M}_{\mathbf{H}} b_j.$$

□

Proposition 30. *The marked cliques satisfy*

$$\mathcal{M}_{(K_{k+1}, (1,\dots,k+1), (1,\dots,k))} = M_k, \qquad \mathcal{M}_{(K_k, (2,\dots,k), (1,\dots,k))} = D_k.$$

Proof. This follows by unfolding the definitions of w and ε. □

Theorem 31. *For every marked NUIG $\mathbf{G}$, $M_{\mathbf{G}} = \mathcal{M}_{\mathbf{G}}$. For $N \in \mathbb{N}$, $M_{\mathbf{G}}^{(N)} = \mathcal{M}_{\mathbf{G}}^{(N)}$, where $\mathcal{M}_{\mathbf{G}}^{(N)}$ is $\mathcal{M}_{\mathbf{G}}$ with j and T restricted to at most N columns and $e_i = e_i(x_1, \dots, x_N)$.*

Proof. As in the proof of Lemma 12, $\mathbf{G}$ is a sum of the marked cliques in Proposition 30, so this follows from Lemma 29. The finite case is proved the same way. □

Taking $a = b = 0$, Theorems 23 and 31 give $X_G(\boldsymbol{x}) = \mathcal{M}_{(G,\emptyset,\emptyset)} = \sum_T w(T \mid \emptyset, G)$, which by Equation (13) gives a new proof of Theorem 26.

On the diagonal of $M_{\mathbf{G}}^{(N)}$, the factors ε cancel. This gives an explicit formula for proper circular arc graphs: unroll the graph as in Lemma 6, place the inputs in columns j, build the rest, and keep the tableaux whose outputs return to the columns j.

Theorem 1. *Let $\Gamma = \operatorname{tr} \mathbf{G}$ be a proper circular arc graph on n vertices with $\mathbf{G} = (G, (n+1, \dots, n+k), (1, \dots, k))$ as in Lemma 6, and let $N \geq n$. Then*

$$X_\Gamma(\boldsymbol{x}) = \sum_j \sum_{\substack{T \supseteq T_j \\ T \setminus T_j = \{k+1, \dots, n+k\} \\ c_T(n+1, \dots, n+k) = j}} \prod_{v=k+1}^{n+k} A\big(c_T(v) \mid c_T(N(v))\big) \frac{e_{c_T(v)}}{e_{c_T(v)-1}},$$

where j ranges over I_k with entries at most N, and T ranges over tableaux with at most N columns.

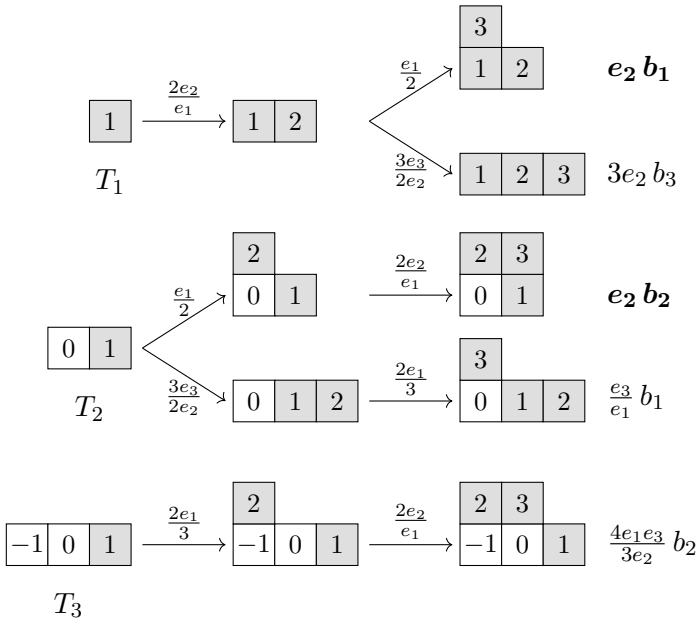
Proof. By Theorems 21 and 31, $X_\Gamma(x_1, \ldots, x_N)$ is the sum of the diagonal entries $w(T \mid T_j, G)$ with $c_T(\alpha) = j$, which is the right side. Since $N \geq n$, both sides have the same expansion in the algebraically independent $e_1, \ldots, e_N$, so the identity holds in infinitely many variables. □

By Equation (13), each summand in Theorem 1 equals

$$\Pr(T \mid T_j, G)\, \frac{\prod_r \lambda_r!\, e_{\lambda_r}}{\prod_r j_r!\, e_{j_r}},$$

where λ is the shape of T, so Theorem 1 is Theorem 26 with the first and last k vertices tied together.

Example 32. Let $\Gamma = P_2$, so $k = 1$ and $\mathbf{G} = (P_3, (3), (1))$ as in Example 17. With $N = 3$, Figure 5 shows every tableau built from T_1, T_2, T_3 with at most 3 columns, with the factor of each added vertex on its arrow. Each final tableau T is labeled by $w(T \mid T_j, G)\, \varepsilon(j)/\varepsilon(c_T(3))\, b_{c_T(3)}$, so summing the labels for each j gives column j of $\mathcal{M}_{\mathbf{G}}^{(3)} = (D_2 M_1)^2$. Its diagonal comes from the two tableaux where vertex 3 returns to column j, so $X_{P_2}(\boldsymbol{x}) = 2e_2$.



$$\mathcal{M}_{\mathbf{G}}^{(3)} = \begin{pmatrix} \boldsymbol{e_2} & \frac{e_3}{e_1} & 0 \\ 0 & \boldsymbol{e_2} & \frac{4e_1e_3}{3e_2} \\ 3e_2 & 0 & \boldsymbol{0} \end{pmatrix}$$

Figure 5: Every build from T_1, T_2, T_3 with at most 3 columns, giving the columns of $\mathcal{M}_{\mathbf{G}}^{(3)} = (D_2 M_1)^2$ for $\mathbf{G} = (P_3, (3), (1))$.

8 Further directions

In this paper, we proved the e-positivity of proper circular arc graphs and provided an alternate proof of the e-positivity of unit interval graphs by proving that color matrices and tableau matrices are related by a change of basis.

Our proof constructs U_k recursively but does not give an explicit formula. In particular, the matrix can be computationally calculated but the formula for each entry becomes very complicated at $k \geq 3$. Finding an explicit formula for U_k and its inverse in finitely many variables would provide a more direct way to calculate tableau matrices and color matrices from one another.

In particular, for any non-NUIG marked graph $\mathbf{G}$ with clique inputs and outputs, Proposition 20 still defines a matrix $M_{\mathbf{G}} = (U_b^{(N)})^{-1} \hat{M}_{\mathbf{G}} U_a^{(N)}$, providing a linear extension of tableau to other graphs. Note $M_{\mathbf{G}}$ need not be built from M_k and D_k, so its entries may not be e-positive. Can these entries be described by a rule on tableaux when the graph is not a NUIG? For example, a rule for attaching a cycle could lead to further e-positivity results for graphs glued along cliques, extending the single-vertex constructions in [21].

Another direction is to seek a q-refinement of these tableau matrices, using the same basis to describe the effect of ascent weights. A corresponding trace formula would extend the q-trace question in [21, Conjecture 74] to gluing along larger cliques. Since the e-coefficients of $X_G(\boldsymbol{x}; q)$ are now known to lie in $\mathbb{N}[q]$ for every natural unit interval graph G [13], the circular case is what remains of the conjecture of Ellzey and of Alexandersson and Panova, and a q-analogue of Theorem 21 would be one approach to it.

Similarly, one can extend the tableau formulas to the Tutte symmetric function in this basis, extending the matrix construction in [22]. Since Tutte weights allow adjacent vertices to have the same color, such an extension would also need to account for repeated colors on the marked vertices. This could perhaps extend Hikita's tableau method to provide a simpler formula for the Tutte symmetric function of NUIGs.

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